

Document Title	Data Analysis SQL Requirements — Olist E-Commerce Project
Reference	QL-OL-002
Project	Olist E-Commerce Analytics
Prepared By	Data & Analytics Division, Quantify Labs
Prepared For	Assigned Analyst / Data Engineer
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1. PURPOSE & SCOPE

This document defines the full analytical requirements for the Olist E-Commerce project engagement. It specifies the business questions that must be answered through SQL, the analytical domains covered, the materialized views to be maintained, and the routine pipeline tasks to be operated via Apache Airflow.

This is a requirements document only. It does not contain SQL code. The assigned analyst is expected to author, test, and submit all corresponding SQL implementations as separate deliverables in accordance with Task Brief QL-OL-001.

2. DATABASE OVERVIEW

The Olist database is a PostgreSQL relational schema representing a Brazilian multi-seller e-commerce marketplace. It captures the complete commercial lifecycle from customer registration and product listing through to order fulfilment, payment, and post-purchase review.

2.1 Core Tables

Table Name	Row Grain	Description
olist_customers	Per customer_id	Customer registration including unique ID, zip code, city and state.
olist_sellers	Per seller_id	Seller registration including location attributes.
olist_orders	Per order_id	Order lifecycle with status, purchase timestamp, approval, carrier and delivery dates.

olist_order_items	Per order-line	Individual line items per order, linked to product and seller, with price and freight.
olist_order_payments	Per payment sequence	One or more payment records per order, covering type, installments and value.
olist_order_reviews	Per review_id	Customer review linked to an order, with numeric score and optional comment.
olist_products	Per product_id	Product catalogue with category, physical dimensions, weight and listing attributes.
olist_product_category_name_translation	Per category	Lookup translating Portuguese category names to English equivalents.
olist_geolocation	Per zip + lat/lng	Geographic coordinates mapped to zip code prefixes for spatial analysis.

2.2 Key Relationships

- Each order belongs to exactly one customer. A customer may have multiple orders identified through customer_unique_id.
- Each order line (order item) references one product and one seller. A single order may span multiple sellers.
- Payment records are linked to orders and may include multiple sequential entries for split or combined payment methods.
- Reviews are linked one-to-one with orders but are not guaranteed to exist for every order.
- Products reference a category name that may be translated via the translation lookup table.

2.3 Schema Notes for the Analyst

- The order_status constraint has been dropped via ALTER TABLE. The analyst must handle all status values present in the data without assuming a fixed enumeration.
- Product dimension columns (weight, length, height, width) have been altered to DECIMAL. Queries must not assume integer behaviour.
- The customer unique ID constraint has been dropped. Multiple customer_id records may share the same customer_unique_id, representing the same physical customer.
- The foreign key between olist_products and the category translation table has been dropped. LEFT JOINS with COALESCE are required to handle unmatched categories.
- Several timestamp columns are nullable. Division by zero and null propagation must be guarded in all date arithmetic queries.

3. ANALYTICAL DOMAINS

All analytical work on this project is organised into five business domains. Every SQL question in Section 4 maps to one of these domains.

#	Domain	Business Focus
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A	Revenue & GMV	Total and segmented revenue, payment patterns, freight contribution, and installment behaviour.
B	Order Fulfilment & Logistics	Delivery performance against estimates, carrier handling times, and late-delivery identification.
C	Customer Behaviour	Purchase frequency, geographic distribution, repeat behaviour, lifetime value, and cohort retention.
D	Seller Performance	Volume, revenue, review scores, on-time delivery rates, and comparative seller ranking.
E	Product Intelligence	Category-level performance, physical attribute correlation with sales, and product co-occurrence patterns.

4. SQL QUESTIONS — SIMPLE TIER

The following eight questions form the foundational analytical layer. They require single or double table joins, basic aggregation, and straightforward filtering. Each must be authored and validated before progressing to the advanced tier.

Q#	Domain	Business Question	Required SQL Features
Q1	A	What is the total number of orders broken down by order status, and what percentage of total orders does each status represent?	GROUP BY, COUNT, window function for percentage of total (SUM OVER).
Q2	A	What is the monthly order volume and total GMV, split between product revenue and freight revenue?	DATE_TRUNC for monthly bucketing, JOIN across orders and order items, SUM aggregation.
Q3	E	Which are the top ten product categories by total revenue, and what is the average item price per category? English category names must be used where available.	Three-table JOIN including translation lookup, LEFT JOIN with COALESCE for unmatched categories, GROUP BY, ORDER BY, LIMIT.
Q4	A	What is the breakdown of payment methods by count, total value, and average number of installments?	Single-table aggregation on olist_order_payments, GROUP BY payment_type, AVG with ROUND.
Q5	E	What is the average review score per product category, limited to categories with more than 50 reviews?	Four-table JOIN, AVG on review_score, HAVING clause to filter low-volume categories.
Q6	C	How are customers distributed across Brazilian states, and how many distinct cities are represented per state?	Single-table aggregation on olist_customers, COUNT DISTINCT for customers and cities, ORDER BY.

Q7	B	Of all delivered orders, what proportion were delivered on time versus late versus missing a delivery date?	CASE WHEN on timestamp comparison, WHERE filter on order_status, COUNT by derived bucket.
Q8	D	Who are the top ten sellers by total revenue, and how many distinct orders did each fulfil?	JOIN of sellers to order items, COUNT DISTINCT on order_id, ORDER BY revenue DESC, LIMIT.

5. SQL QUESTIONS — ADVANCED TIER

The following nine questions require advanced SQL techniques including common table expressions, window functions, ordered-set aggregates, self-joins, and conditional subqueries. They may only be attempted once the simple tier has passed review.

Q#	Domain	Business Question	Required SQL Features
Q9	C	What is the lifetime value of each customer, total orders placed, and are they a one-time or repeat purchaser? Aggregate to show averages by customer type.	Multi-level CTE, GROUP BY customer_unique_id, CASE WHEN for customer type classification, outer aggregate across the CTE result.
Q10	A	What is the running total GMV by month, and what is the month-over-month percentage growth rate?	CTE for monthly base aggregation, LAG window function for prior month comparison, SUM OVER for running total, NULLIF to guard zero division.
Q11	D	For each seller with at least 20 delivered orders, what is the average, median, 90th and 95th percentile delivery delay in days relative to the estimated delivery date?	CTE for delay calculation using EXTRACT and EPOCH, PERCENTILE_CONT ordered-set aggregate, HAVING for minimum volume threshold.
Q12	C	By monthly acquisition cohort, what percentage of customers placed a second order within 90 days of their first purchase?	Three-level CTE, EXISTS correlated subquery to detect return purchases within a date window, DATE_TRUNC for cohort grouping, percentage calculation.
Q13	B	How does average review score vary across delivery delay buckets — very early, on time, slightly late, moderately late, and very late?	CTE for delay computation, CASE WHEN bucketing on delay days, JOIN to reviews, AVG score and negative review count per bucket.
Q14	D	For all qualifying sellers, assign a revenue rank, a review score rank, and classify each into revenue and review quartiles.	CTE for seller metric aggregation, DENSE_RANK and NTILE window functions partitioned over the full result set.

Q1 5	E	Which pairs of products are most frequently purchased within the same order, and how often does each pairing occur?	Self-join on olist_order_items using order_id with an inequality condition on product_id to avoid duplicates, COUNT for co-occurrence frequency.
Q1 6	A	Which orders used more than one payment method, and what is the combined payment string, total value paid, and maximum installment count for each?	STRING_AGG with ORDER BY inside aggregation, GROUP BY order_id, HAVING COUNT greater than one for multi-payment filtering.
Q1 7	B / E	By product category and customer state, what is the average freight value as a percentage of product price for orders with sufficient volume?	Five-table JOIN chain, NULLIF to guard division by price, AVG of the computed ratio, HAVING for minimum item threshold.

6. SQL FEATURE COVERAGE REQUIREMENTS

The combined set of 17 queries must collectively demonstrate competency across all features listed below. The final submission will be assessed against this checklist during Stage 2 review.

SQL Feature	Demonstrated In
Basic aggregation — COUNT, SUM, AVG, ROUND	Q1, Q2, Q3, Q4, Q6
HAVING clause for post-aggregation filtering	Q5, Q11, Q15, Q17
CASE WHEN conditional logic	Q7, Q9, Q13, Q14
Multi-table JOIN chains (3 or more tables)	Q5, Q13, Q17
LEFT JOIN with COALESCE for nullable lookups	Q3, Q5, Q17
DATE_TRUNC for time-period bucketing	Q2, Q10, Q12
EXTRACT and EPOCH for interval arithmetic	Q11, Q13
Common Table Expressions (WITH clauses)	Q9, Q10, Q11, Q12, Q13, Q14
LAG window function for period-over-period comparison	Q10
SUM OVER running total window	Q1, Q10
DENSE_RANK and NTILE for ranking and quartiles	Q14
PERCENTILE_CONT ordered-set aggregate	Q11
EXISTS correlated subquery	Q12
Self-join with inequality predicate	Q15

STRING_AGG ordered string concatenation	Q16
NULLIF to prevent zero-division errors	Q10, Q17

7. MATERIALIZED VIEW REQUIREMENTS

Four materialized views must be created and maintained as the pre-computed analytical layer. Each view must be accompanied by a UNIQUE index to enable CONCURRENTLY refresh, which prevents table locking during scheduled updates.

MV1 — Daily Order Summary

View Name	mv_daily_order_summary
Refresh	Daily — 02:00 UTC
Unique Index	Composite on (order_date, order_status)
Purpose	Pre-aggregates order count, unique customer count, product revenue, freight revenue, and total GMV by day and order status. Serves daily KPI dashboards and Airflow sink tasks.
Source Tables	olist_orders, olist_order_items

MV2 — Seller Performance Snapshot

View Name	mv_seller_performance
Refresh	Daily — 02:30 UTC
Unique Index	On seller_id
Purpose	Aggregates per-seller metrics including total orders fulfilled, total revenue, average review score, review volume, average delivery delay in days, and count of on-time deliveries. Filters to delivered orders with a non-null delivery date.
Source Tables	olist_sellers, olist_order_items, olist_orders, olist_order_reviews

MV3 — Category Revenue by Month

View Name	mv_category_monthly_revenue
Refresh	Weekly — Monday 03:00 UTC
Unique Index	Composite on (month, category)

Purpose	Aggregates order count, total revenue, average item price, and average review score per product category per calendar month. English category names must be resolved via the translation lookup with COALESCE fallback.
Source Tables	olist_orders, olist_order_items, olist_products, olist_product_category_name_translation, olist_order_reviews

MV4 — Customer RFM Scores

View Name	mv_customer_rfm
Refresh	Weekly — Monday 03:30 UTC
Unique Index	On customer_unique_id
Purpose	Computes Recency (days since last order), Frequency (total delivered orders), and Monetary (total spend including freight) per unique customer. Assigns NTILE(5) scores for each dimension to enable RFM-based segmentation in downstream tasks.
Source Tables	olist_customers, olist_orders, olist_order_items

8. AIRFLOW PIPELINE REQUIREMENTS

Four Airflow DAGs must be implemented to automate scheduled refresh, aggregation, and data quality monitoring. All DAGs must be idempotent and include retry logic. Task dependency chains must be explicit — no parallel execution unless explicitly approved.

DAG 1 — olist_daily_refresh | Schedule: 02:00 UTC Daily

Task ID	Description	Operator Type
check_new_data	Wait for records with today's order_purchase_timestamp before proceeding.	SQLSensor
refresh_mv_daily_orders	Refresh mv_daily_order_summary using CONCURRENTLY to avoid locking.	PostgresOperator
refresh_mv_seller_performance	Refresh mv_seller_performance using CONCURRENTLY.	PostgresOperator
compute_daily_kpis	Upsert yesterday's aggregated KPIs (orders, GMV, new customers) into the agg_daily_kpis sink table using ON CONFLICT DO UPDATE.	PostgresOperator
alert_late_deliveries	Identify delivered orders where actual delivery exceeds estimate by more than 3 days and write results to alerts_late_delivery.	PostgresOperator

notify_slack	Post daily KPI summary (orders, GMV, alert count) to the #data-ops Slack channel.	SlackWebhookOperator
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DAG 2 — olist_weekly_aggregates | Schedule: Monday 03:00 UTC

Task ID	Description	Operator Type
refresh_mv_category_revenue	Refresh mv_category_monthly_revenue CONCURRENTLY.	PostgresOperator
refresh_mv_customer_rfm	Refresh mv_customer_rfm CONCURRENTLY.	PostgresOperator
compute_weekly_seller_ranking	Compute per-seller revenue and review rank for the week and upsert into agg_seller_weekly_rank.	PostgresOperator
compute_cohort_retention	Run the 90-day cohort retention calculation and upsert results into agg_cohort_retention by cohort month.	PostgresOperator
flag_churned_customers	Tag customers with recency exceeding 180 days as churned in the agg_customer_segments table.	PostgresOperator
compute_payment_mix	Aggregate weekly payment type splits and upsert into agg_payment_weekly.	PostgresOperator

DAG 3 — olist_monthly_reports | Schedule: 1st of Month 04:00 UTC

Task ID	Description	Operator Type
full_refresh_all_mvs	Sequentially refresh all four materialized views to ensure full alignment at month-end.	PostgresOperator
compute_monthly_clv	Run the customer lifetime value calculation and upsert into agg_customer_clv_monthly.	PostgresOperator
category_performance_report	Export the prior month's category P and L snapshot into reports.category_monthly.	PostgresOperator
seller_scorecard	Produce a multi-metric seller scorecard with NTILE quartile bands across revenue, review score, and on-time rate.	PostgresOperator
freight_cost_analysis	Compute freight as a percentage of product price segmented by category and state and upsert into agg_freight_analysis.	PostgresOperator
export_to_s3	Export all aggregate sink tables to S3 as Parquet files for consumption by the BI layer.	PythonOperator

DAG 4 — olist_data_quality | Schedule: Every 6 Hours

Task ID	Check Description	On Failure	Severity
check_orphan_order_items	Detect order items with no matching record in olist_orders via a LEFT JOIN null check.	Raise alert	CRITICAL
check_negative_prices	Count any order items where price or freight_value is below zero.	Raise alert	CRITICAL
check_review_score_range	Identify review records where review_score falls outside the range of 1 to 5 inclusive.	Raise alert	HIGH
check_future_purchase_dates	Flag orders where order_purchase_timestamp exceeds the current timestamp.	Raise alert	HIGH
check_delivered_missing_date	Count delivered orders with a NULL order_delivered_customer_date.	Log warning	MEDIUM
check_payment_value_zero	Identify payment records where payment_value is exactly zero.	Log warning	MEDIUM

9. AGGREGATE SINK TABLE REQUIREMENTS

The Airflow pipeline writes computed results into the following aggregate sink tables. The analyst must author DDL for each table with appropriate primary keys, data types, and ON CONFLICT targets.

Table Name	Primary Key	Written By DAG
agg_daily_kpis	kpi_date	DAG 1 — compute_daily_kpis
agg_seller_weekly_rank	Composite: week_start, seller_id	DAG 2 — compute_weekly_seller_ranking
agg_customer_segments	customer_unique_id	DAG 2 — flag_churned_customers
agg_cohort_retention	Composite: cohort_month, retention_window_days	DAG 2 — compute_cohort_retention
agg_payment_weekly	Composite: week_start, payment_type	DAG 2 — compute_payment_mix
agg_customer_clv_monthly	Composite: report_month, customer_unique_id	DAG 3 — compute_monthly_clv
agg_freight_analysis	Composite: month, category, customer_state	DAG 3 — freight_cost_analysis
alerts_late_delivery	Composite: alert_date, order_id	DAG 1 — alert_late_deliveries

10. SUBMISSION STANDARDS

All work submitted against this requirements document must comply with the Quantify Labs SQL standards published in Task Brief QL-OL-001. The following are mandatory for acceptance.

SQL Standards

- PostgreSQL-compatible syntax throughout. No vendor-specific functions outside the PostgreSQL dialect.
- All CTEs must be used in preference to nested subqueries wherever logic exceeds two levels.
- Column aliases in snake_case. Descriptive names required — no single-letter aliases outside self-joins.
- Every query must open with a single-line comment stating the business question being answered.
- NULLIF required on all division operations. COALESCE required on all nullable foreign-key lookups.
- Window functions must include an explicit ORDER BY inside the OVER clause.

Airflow Standards

- All DAGs must define default_args with retries set to 2 or higher and a retry_delay of 5 minutes.
- All SQL executed via Airflow that writes to sink tables must be idempotent using ON CONFLICT DO UPDATE.
- Task dependency chains must be declared explicitly with the >> operator. No implicit ordering.
- Each DAG file must include a docstring describing its schedule, purpose, and owner.

Review Sequence

1. Submit completed simple-tier queries (Q1–Q8) for Stage 1 peer review.
2. On clearance, submit advanced-tier queries (Q9–Q17) for Stage 1 peer review.
3. Submit all four materialized view DDL scripts with index statements for Stage 1 review.
4. Submit all four Airflow DAG files and all aggregate sink DDL for Stage 1 review.
5. On full Stage 1 clearance, full deliverable package proceeds to Stage 2 lead review.

DOCUMENT SIGN-OFF

Approved By — Quantify Labs	Acknowledged By — Analyst
Name: Quantify labs CEO Role: Team Lead, Data & Analytics Date: 23 May 2026	Name: _____ Role: Data Analyst / Data Engineer Date: _____